

DATA

SMS Spam Collection dataset
5,574 emails · from Kaggle (UCI)

CLEAN

Data cleaning

Drop junk columns · rename v1→target, v2→text · remove duplicates

Label encoding

ham → 0 · spam → 1

NLP

NLP text preprocessing pipeline (applied to every email)

Lowercase

"Hello" → "hello"

Tokenize

split into words

Remove stopwords

"the", "is", "a" gone

Porter Stemmer

"dancing" → "danc"

TF-IDF Vectorizer

top 3,000 features · text → numbers

MODEL

Model comparison (3 tried)

Gaussian NB

rejected · low precision

Multinomial NB

rejected · lower score

Bernoulli NB ✓

chosen · best precision

DEPLOY

Save with pickle

model1.pkl · vectorizer2.pkl

Flask (main.py)

receives form POST

transform_text()

same NLP pipeline above

Predict + render

SPAM or NOT SPAM

User sees verdict in browser